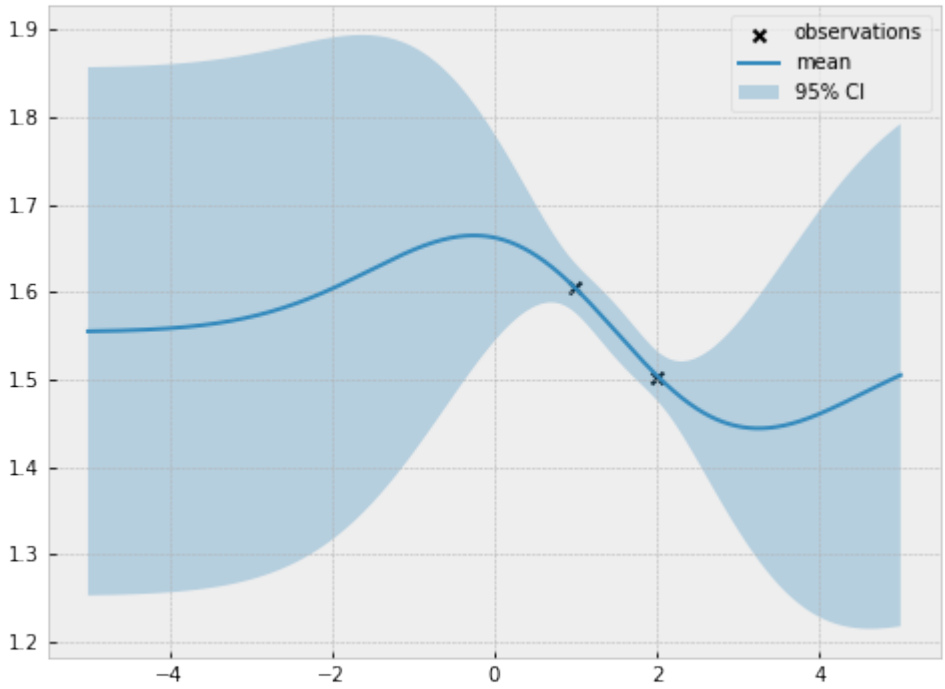


Figure 4.6 Predictions by a GP trained on two data points from the Forrester function.

Each BayesOpt policy has a different way of addressing the tradeoff and therefore offers different advice regarding how to best explore the search space. In Figure 4.6, some policies might lead us to further explore unknown regions, while others might suggest zeroing in on what we know to give high values. Again, there's usually no one-size-fits-all approach (that is, no policy that always works well).

4.2 Finding improvement in Bayesian optimization

We almost have everything ready to run BayesOpt on a given objective function. We now need is a policy that gives a scoring rule about how valuable each data point that we could potentially label is in our search for the optimum of the objective. Again, each policy that we will see gives a different scoring rule, motivated by different heuristics for optimization.

In this section, we will learn about one such heuristic that makes intuitive sense when the goal is optimization. It comes in the form of seeking to improve from the best point that we have seen so far throughout optimization. In the upcoming subsection, we will learn that GPs are amenable to facilitating the computation of this measure of improvement. Then, we will see how different ways to defining improvement give rise to two of the most common BayesOpt policies: *Probability of Improvement* and *Expected Improvement*.

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